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Command Prompt - mysql -u x + v
Microsoft Windows [Version 10.0.26200.8875]
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C:\Users\Yeshvitha>cd C:\Program Files\MySQL\MySQL Server 8.0\bin

C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 14
Server version: 8.0.46 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> create database libflow;
Query OK, 1 row affected (0.04 sec)

mysql> use libflow;
Database changed
mysql> CREATE TABLE Books ( book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(20) UNIQUE, published_year INT CHECK (
published_year < 2027) );
Query OK, 0 rows affected (0.09 sec)

mysql> INSERT INTO Books (book_id, title, isbn, published_year) VALUES (1, 'The Great Gatsby', '9780743273565', 1925), (2, 'To Kill a
Mockingbird', '9780061120084', 1960), (3, '1984', '9780451524935', 1949); SELECT * FROM Books;
Query OK, 3 rows affected (0.03 sec)
Records: 3 Duplicates: 0 Warnings: 0

+-----+-----+-----+-----+
| book_id | title           | isbn           | published_year |
+-----+-----+-----+-----+
| 1       | The Great Gatsby | 9780743273565 | 1925           |
+-----+-----+-----+-----+
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+-----+-----+-----+-----+
| book_id | title           | isbn           | published_year |
+-----+-----+-----+-----+
| 1       | The Great Gatsby | 9780743273565 | 1925           |
| 2       | To Kill a Mockingbird | 9780061120084 | 1960           |
| 3       | 1984            | 9780451524935 | 1949           |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> REATE TABLE Members ( member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR(100) UNIQUE ); INSERT INTO Members (mem
ber_id, full_name, email) VALUES (101, 'John Smith', 'john.smith@email.com'), (102, 'Emma Wilson', 'emma.wilson@email.com'), (103, 'M
ichael Brown', 'michael.brown@email.com'); SELECT * FROM Members;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the righ
t syntax to use near 'REATE TABLE Members ( member_id INT PRIMARY KEY, full_name VARCHAR(100), email V' at line 1
ERROR 1146 (42502): Table 'libflow.members' doesn't exist
ERROR 1146 (42502): Table 'libflow.members' doesn't exist
mysql> Create TABLE Members ( member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR(100) UNIQUE ); INSERT INTO Members (m
ember_id, full_name, email) VALUES (101, 'John Smith', 'john.smith@email.com'), (102, 'Emma Wilson', 'emma.wilson@email.com'), (103,
'Michael Brown', 'michael.brown@email.com'); SELECT * FROM Members;
Query OK, 0 rows affected (0.10 sec)

Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

+-----+-----+-----+
| member_id | full_name      | email          |
+-----+-----+-----+
| 101       | John Smith     | john.smith@email.com |
| 102       | Emma Wilson    | emma.wilson@email.com |
| 103       | Michael Brown  | michael.brown@email.com |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> CREATE TABLE Loans ( loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, FOREIGN KEY (member_id) REFERENCES M
embers(member_id), FOREIGN KEY (book_id) REFERENCES Books(book_id) );
Query OK, 0 rows affected (0.12 sec)
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mysql> INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES (1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'), (3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'), (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'), (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'), (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01'); SELECT * FROM Loans;
Query OK, 10 rows affected (0.03 sec)
Records: 10 Duplicates: 0 Warnings: 0
```

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01
8	102	3	2025-03-07
9	103	1	2025-03-15
10	101	1	2025-04-01

```
10 rows in set (0.00 sec)

mysql> SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Members m ON l.member_id = m.member_id INNER JOIN Books b ON l.book_id = b.book_id;
```

Member_Name	Book_Title
John Smith	The Great Gatsby
John Smith	To Kill a Mockingbird
John Smith	1984
John Smith	The Great Gatsby
Emma Wilson	To Kill a Mockingbird
Emma Wilson	The Great Gatsby
Emma Wilson	1984
Michael Brown	1984
Michael Brown	To Kill a Mockingbird

24°C Light rain 10:14 AM 30-07-2026

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| Michael Brown | 1984 |
| Michael Brown | To Kill a Mockingbird |
| Michael Brown | The Great Gatsby |
+-----+-----+
10 rows in set (0.00 sec)

mysql> SELECT m.full_name AS Member_Name, b.title AS Book_Title
-> FROM Loans l
-> INNER JOIN Members m ON l.member_id = m.member_id
-> INNER JOIN Books b ON l.book_id = b.book_id;
```

Member_Name	Book_Title
John Smith	The Great Gatsby
John Smith	To Kill a Mockingbird
John Smith	1984
John Smith	The Great Gatsby
Emma Wilson	To Kill a Mockingbird
Emma Wilson	The Great Gatsby
Emma Wilson	1984
Michael Brown	1984
Michael Brown	To Kill a Mockingbird
Michael Brown	The Great Gatsby

```
10 rows in set (0.00 sec)

mysql> SELECT published_year, COUNT(book_id) AS Total_Books
-> FROM Books
-> GROUP BY published_year
-> ORDER BY published_year;
```

published_year	Total_Books
1925	1
1949	1
1960	1

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+-----+
3 rows in set (0.03 sec)

mysql> CREATE TABLE Donation_History (
->   donation_id INT PRIMARY KEY,
->   book_id INT,
->   donor_name VARCHAR(100),
->   donation_date DATE,
->   FOREIGN KEY (book_id) REFERENCES Books(book_id)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO Books (book_id, title, isbn, published_year)
-> VALUES (4, 'Animal Farm', '9780451526342', 1945);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
-> VALUES (1, 4, 'Raj Kumar', CURDATE());
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE INDEX idx_books_isbn ON Books(isbn);
Query OK, 0 rows affected (0.08 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Books WHERE isbn = '9780451524935';
+-----+
| book_id | title | isbn | published_year |
+-----+
| 3 | 1984 | 9780451524935 | 1949 |
+-----+
```

```
mysql> SELECT * FROM Books WHERE isbn = '9780451524935';
+-----+
| book_id | title | isbn | published_year |
+-----+
| 3 | 1984 | 9780451524935 | 1949 |
+-----+
1 row in set (0.00 sec)

mysql> |
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